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Question **1**

Complete

Not graded

If the series $\sum_{n=1}^{\infty}(a_n + b_n)$ diverges, which of the following must be true?

Select one:

- a. $\sum_{n=1}^{\infty} a_n$ converges.
- b. $\sum_{n=1}^{\infty} b_n$ diverges.
- c. $\lim_{n\to\infty}(a_n + b_n) \neq 0$
- d. $\sum_{n=1}^{\infty} \frac{a_n+b_n}{2}$ diverges.

$$\sum_{n=1}^{\infty}\frac{a_n + b_n}{2} = \frac{1}{2}\sum_{n=1}^{\infty}(a_n + b_n)$$

Because $\sum_{n=1}^{\infty}(a_n + b_n)$ diverges,

$\frac{1}{2}\sum_{n=1}^{\infty}(a_n + b_n)$ must be divergent.

The correct answer is: $\sum_{n=1}^{\infty} \frac{a_n+b_n}{2}$ diverges.

Question **2**

Complete

Not graded

Evaluate $\sum_{n=1}^{\infty} 5^{-n} - 7^{-n}$.

Select one:

- a. 0
- b. $2/3$
- c. $1/12$
- d. ∞

$$\sum_{n=1}^{\infty} 5^{-n} - 7^{-n} = \sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^n - \sum_{n=1}^{\infty} \left(\frac{1}{7}\right)^n$$

Each of these is a convergent geometric series with $r = 1/5$ and $r = 1/7$, respectively.

The correct answer is: $1/12$

Question **3**

Complete

Not graded

If $\lim_{n\to\infty} a_n b_n = 0$, which of the following is true?

Select one:

- a. $\sum_{n=1}^{\infty} a_n$ converges
- b. $\sum_{n=1}^{\infty} a_n b_n$ converges
- c. $\sum_{n=1}^{\infty} b_n$ converges
- d. None of the above

Answers A, B, and C are all incorrect.

The correct answer is: None of the above

Question **4**

Complete

Not graded

Evaluate \(\sum_{n = 1}^{\infty}{\frac{1+5^n+3^n}{7^n}} \).

Select one:

- a. \(0 \)
- b. \(\infty \)
- c. \(41/12 \)
- d. \(259/120 \)

$$\sum_{n=1}^{\infty}\frac{1+5^n+3^n}{7^n}=\sum_{n=1}^{\infty}\left(\frac{1}{7}\right)^n+\sum_{n=1}^{\infty}\left(\frac{5}{7}\right)^n+\sum_{n=1}^{\infty}\left(\frac{3}{7}\right)^n$$

Each of these is a convergent geometric series. Because the first index in each series is $n=1$, each sum can be computed using the formula,

$$\sum_{n=1}^{\infty}ar^n=\sum_{n=0}^{\infty}ar^n-a=\frac{a}{1-r}-a=\frac{ar}{1-r}.$$

Thus,

$$\begin{aligned}\sum_{n=1}^{\infty}\frac{1+5^n+3^n}{7^n}&=\sum_{n=1}^{\infty}\left(\frac{1}{7}\right)^n+\sum_{n=1}^{\infty}\left(\frac{5}{7}\right)^n+\sum_{n=1}^{\infty}\left(\frac{3}{7}\right)^n\\&=\frac{1/7}{6/7}+\frac{5/7}{2/7}+\frac{3/7}{4/7}\\&=\frac{1}{6}+\frac{5}{2}+\frac{3}{4}\\&=\frac{2+30+9}{12}\\&=\frac{41}{12}\end{aligned}$$

The correct answer is: \(41/12 \)

Question **5**

Complete

Not graded

Evaluate the following as true or false.
The series \(\sum_{n=1}^{\infty}{a_n\,^{100}} \) must diverge.

Select one:

- True
- False

This statement is not true,because
$\lim_{n\to\infty} a_n^{100}$ might be 0 (if $\lim_{n\to\infty} a_n = 0$).
Therefore, you do not know whether
this series diverges or not at
this stage.

The correct answer is 'False'.

Question **6**

Complete

Not graded

Suppose \(\sum_{n=1}^{\infty}{a_n} \) converges and \(\lim_{n \to \infty} b_n \neq 0 \). Which of the following is true?

Select one:

- a. \(\sum_{n=1}^{\infty}{\frac{a_n}{b_n}} \) converges.
- b. \(\sum_{n=1}^{\infty}{a_n\,^2} \) converges.
- c. \(\sum_{n=1}^{\infty}{(3+a_n)} \) converges.
- d. \(\sum_{n = 1}^{\infty}{a_n}+\sum_{n=1}^{\infty}{b_n} \) diverges.

Because $\lim_{n\to\infty} a_n = 0$ and $\lim_{n\to\infty} b_n \neq 0$,
you know that $\lim_{n\to\infty} (a_n + b_n) \neq 0$.
Accordingly, $\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$ diverges.

The correct answer is: \(\sum_{n = 1}^{\infty}{a_n}+\sum_{n=1}^{\infty}{b_n} \) diverges.

Question **7**

Complete

Not graded

Evaluate the following as true or false:

If \(\sum_{n=1}^{\infty}{a_n} \) and \(\sum_{n=1}^{\infty}{b_n} \) both converge, we can conclude that \(\sum_{n=1}^{\infty}{a_nb_n} \) converges.

Select one:

- True
- False

You can conclude from the given condition only that $\lim_{n\to\infty} a_n b_n = 0$. This does not guarantee the convergence of $\sum_{n=1}^{\infty} a_n b_n$.

The correct answer is 'False'.

Question **8**

Complete

Not graded

Which of the following series converges?

Select one:

- a. \(\sum_{n = 1}^{\infty}{\frac{n^2-1}{3n^2+1}} \)
- b. \(\sum_{n = 1}^{\infty}{\frac{1}{1+\bigg(\frac{9}{10}\bigg)^n}} \)
- c. \(\sum_{n = 1}^{\infty}{\bigg(\frac{2}{n}-\frac{1}{2^n}\bigg)} \)
- d. \(\sum_{n =1 }^{\infty}{\frac{99^n}{100^n}} \)

This is a convergent geometric series with $r = \frac{99}{100}$.

The correct answer is: \(\sum_{n =1 }^{\infty}{\frac{99^n}{100^n}} \)

Question **9**

Complete

Not graded

If \(b_n \neq 0 \), which of the following statements must be true?

Select one:

- a. \(\sum_{n = 1}^{\infty}{b_n} \) diverges.
- b. \(\sum_{n = 1}^{\infty}{b_n} \) converges.
- c. \(\sum_{n = 1}^{\infty}{b_n\,^{100}} \) diverges.
- d. None of the above.

None of the answers are correct.

The correct answer is: None of the above.

Question **10**

Complete

Not graded

Evaluate \(\sum_{n = 1}^{\infty}{\frac{5^{n+2}}{7^n}} \).

Select one:

- a. \(5/2 \)
- b. \(0 \)
- c. \(\infty \)
- d. \(125/2 \)

$$\sum_{n=1}^{\infty}\frac{5^{n+2}}{7^n}=\sum_{n=1}^{\infty}\frac{5^n\cdot 5^2}{7^n}$$

$$=\sum_{n=1}^{\infty}25\frac{5^n}{7^n}$$

$$=25\cdot\sum_{n=1}^{\infty}\left(\frac{5}{7}\right)^n$$

This results in $\frac{125}{2}$.

The correct answer is: \(125/2 \)

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